BLOCKCHAIN LAB 03

Consensus Simulation and Bonus Challenges

Proof of Work · Casper FFG · Gossip networking · Double-spend · Scalable mining · Selfish mining

Submission information	Details
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Repository	https://github.com/korneliasara/blockchain-lab03-consensus-simulation
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Executive summary. This project implements three teaching-scale blockchain consensus subsystems and evaluates three optional bonus challenges. The recorded final checks show 57/57 local tests and 42/42 professor reference tests passing. Bonus E2 adds 5/5 passing tests, and E3 adds 11/11 passing tests. Test passes are evidence of correctness against these suites, not a guarantee of grading credit.

Evidence note. Numerical results are transcribed from the experiment screenshots shared during development. The three bonus figures in this report are freshly plotted from the recorded numbers or analytical formulas, and labelled as such. Original plots, raw CSVs and source code are linked in the repository; their image files were not directly available for embedding here.

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1 Core implementation and verification

The Python project includes consensus/chain.py, pow.py, pos.py, network.py and CLI support. State transitions, chain fork choice, proof-of-work validation, validator accounting and gossip behavior were developed and checked incrementally.

Suite	Verified result	Notes
Local Lab 03 tests	57 passed	test_consensus.py
Professor reference tests	42 passed	professor_tests/test_reference.py (not included in repository)
Bonus E2 tests	5 passed	Persistent worker/valid nonce checks
Bonus E3 tests	11 passed	Threshold, model and simulation checks

The core results were checked again after the bonus implementations were added. The independent bonus modules are stored under bonus_challenges/, separate from the compulsory consensus/ package.

2 Part A - Proof of Work

The implemented mechanisms include deterministic block hashing, transaction/state validity, nonce search, difficulty-target conversion and retargeting, parallel nonce search and cumulative-work fork choice. A chain with greater accumulated work is preferred rather than simply the highest block count.

100-block timing experiment

Metric	Recorded value
Target interval	1.000 s
Mean observed interval	0.7611 s
Median observed interval	0.5157 s
Standard deviation	0.8348 s
Median / mean	0.678
Standard deviation / mean	1.097

The observed interval distribution is strongly right-skewed, compatible with the memoryless waiting-time intuition of repeated independent hash attempts. Difficulty retargeting changes the expected time between blocks, so the aggregate histogram is not a pure fixed-rate exponential sample.

Original figures: block interval histogram and difficulty-adjustment plot.

Parallel demonstration

The earlier demonstration found a valid nonce with 1, 2 and 4 workers, with individual observed times of 0.2671 s, 0.2893 s and 0.2049 s, respectively. Those three observations alone do not establish stable scaling; the dedicated E2 benchmark addresses this separately.

3 Part B - Proof of Stake and Casper FFG

The simplified model includes validator effective balances, attestations, checkpoint justification and finalisation, double-vote and surround-vote detection, slashing and inactivity penalties.

Checkpoint finality

Epoch	Newly justified	Newly finalised
1	Checkpoint 1	None
2	Checkpoint 2	Checkpoint 1
3	Checkpoint 3	Checkpoint 2

The demo exhibits the required two-step progression: one justified checkpoint is not sufficient on its own; subsequent justified links can finalise earlier checkpoints.

Correlated slashing

Simultaneous offenders	Loss per validator
1	2.4084 ETH
5	8.0558 ETH
10	15.1462 ETH
20	29.4320 ETH
30	32.0000 ETH

The simulated penalty increases with the scale of coordinated misbehavior, reaching the modelled maximum stake loss of 32 ETH at 30 offenders. Original slashing figure.

Inactivity leak and recovery

A simulation begins with 24 of 64 validators participating (37.5% of stake). As inactive validators lose effective stake, the active share rises to 66.86% at epoch 82 and approximately 67.21% by epoch 83. The recorded results show justification recovering at epoch 82 and finalisation resuming by epoch 83. This demonstrates a simplified liveness recovery after a prolonged participation failure.

Original inactivity-leak figure.

4 Part C - Gossip networking and partition recovery

The gossip implementation exercises asynchronous propagation, deduplication, transaction validity, simulated network partitioning, reconnection and chain reorganisation.

Stage	Group A: nodes 0,1,2	Group B: nodes 3,4,5
Isolated	Height 3; work 262144	Height 5; work 393216
After reconnection	Height 5; one reorg per node; depth 3	Height 5; no reorg

After healing, all six nodes agreed on the same winning chain at height 5. The heavier branch prevailed. The in-process simulation validates reconciliation logic, but it is not equivalent to a deployed peer-to-peer TCP network with independently verified remote blocks.

5 Bonus E1 - Double-spend probability (+1 possible point)

The analytical calculation implements the Poisson catch-up approximation from Section 11 of the Bitcoin whitepaper. Monte Carlo runs draw from that same approximation and sample the eventual catch-up event. Consequently, agreement checks the implementation of the stated model, not an independently modelled full mining network.

Attacker share q	Confirmations z	Analytical	Simulated	Relative error
0.10	5	0.0009137	0.0008820	3.47%
0.10	6	0.0002428	0.0002288	5.77%
0.30	5	0.1773523	0.1774250	0.04%

Monte Carlo sizes were 1,000,000 for the first and third cases, and 5,000,000 for q=0.10, z=6. All three recorded relative errors were within the assignment's approximate 10% target.

Attacker share	Below 10%	Below 1%	Below 0.1%
0.10	2	4	5
0.30	7	16	24
0.45	102	220	340

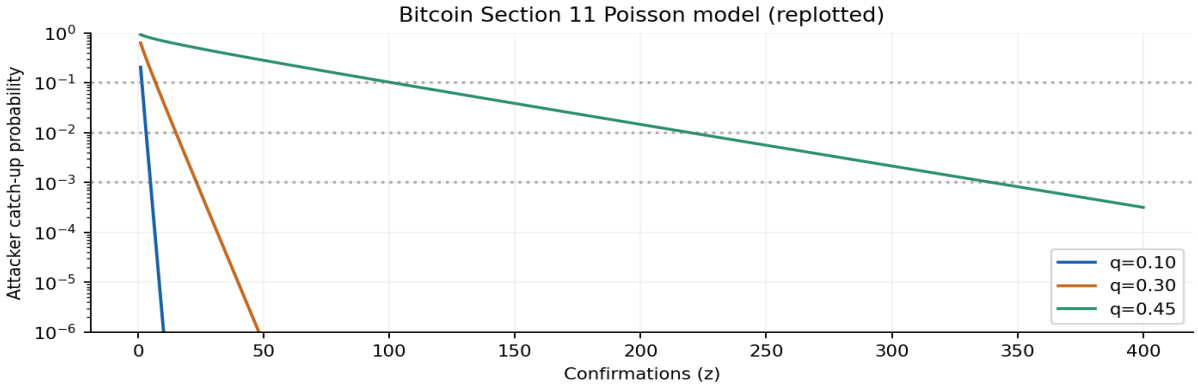


Figure 1. Analytical Section 11 probabilities freshly replotted from the implemented formula. Original experiment figure and threshold CSV are linked in the E1 folder.

E1 files: source, Monte Carlo, data, README and original graph.

6 Bonus E2 - Reusable scalable miner (+2 possible points)

The separate E2 implementation creates persistent Windows-compatible multiprocessing workers once, sends new headers and targets through job queues, searches disjoint nonce sequences, coordinates termination and verifies the winning nonce. A diagnostic confirms that worker PIDs remain unchanged across jobs; five focused tests pass.

Final selected 100-block benchmark

Statistic	1 worker	2 workers	4 workers
Mean time (s)	0.13963	0.06425	0.03622
Median / mean	0.634	0.600	0.555
Std dev / mean	0.952	1.159	1.700
Speed-up	1.00x	2.17x	3.86x

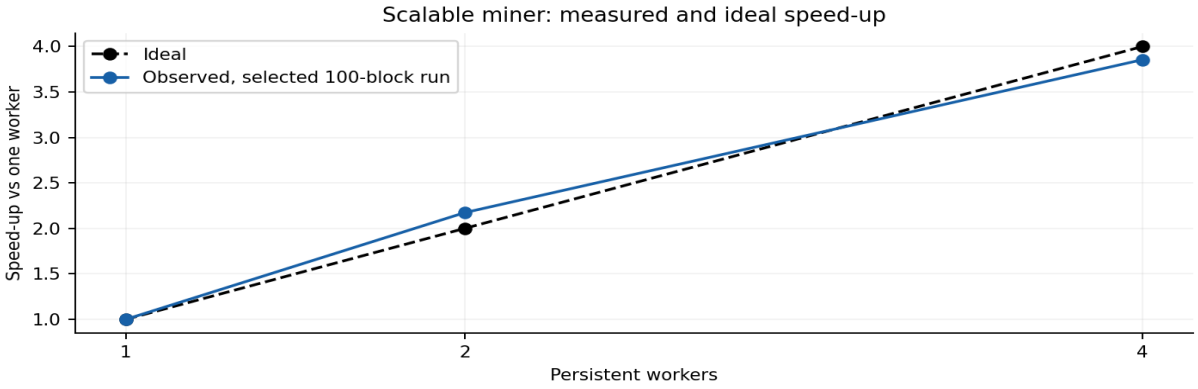


Figure 2. Measured speed-up replotted from the recorded E2 summary, alongside the ideal linear reference.

The selected run gives mean-time ratios 1 : 0.46 : 0.26, close to the target 1 : 0.5 : 0.25. However, its four-worker standard-deviation/mean ratio of 1.700 is noticeably above the ideal exponential value of 1.0. Earlier runs also showed poor four-worker scaling. Scheduling, coordination overhead, stochastic variation and finite sample size can affect observed results; the fastest run should not be presented as universally representative.

Scope qualification: E2 currently benchmarks a standalone SHA-256 nonce/header function; it is not a drop-in queue-based rewrite of the original consensus.pow miner. This matters because the wording asks to rewrite the parallel miner, and the professor may require direct integration to award the full bonus.

E2 files: worker code, diagnostic, automated tests, data and charts.

7 Bonus E3 - Eyal-Sirer selfish mining (+2 possible points)

This standalone event simulation models the private-chain lead, attacker withholding and selective publication, public ties and allocation of accepted block rewards. The attacker's total hash share is alpha; gamma denotes the proportion of honest hash power that follows the attacker's branch in a tie.

The theoretical profitability boundary is alpha > (1 - gamma)/(3 - 2 gamma), yielding alpha > 1/3 when gamma=0 and alpha > 1/4 when gamma=0.5.

alpha	gamma	Honest share	Theoretical selfish	Simulated selfish
0.20	0	20.00%	12.97%	12.96%
0.30	0	30.00%	27.31%	27.25%
0.40	0	40.00%	48.37%	48.31%
0.20	0.5	20.00%	18.24%	18.19%
0.30	0.5	30.00%	32.69%	32.61%
0.40	0.5	40.00%	52.56%	52.49%

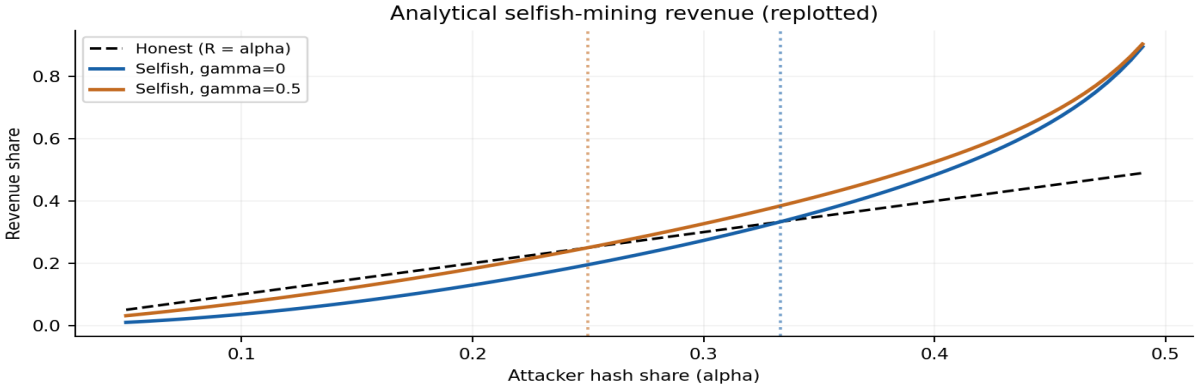


Figure 3. Eyal-Sirer analytical curves freshly replotted for gamma=0 and 0.5; simulated values appear in the table above. See the E3 repository for the original theory-versus-simulation plot.

The attack illustrates that a minority pool may obtain a reward share exceeding its hash-power share through strategic withholding. It does not show that such a pool can automatically perform arbitrary majority-chain attacks or reverse any transaction. The experiment's 11 E3 automated tests passed.

E3 files: implementation, tests, CSV, original plot and README.

8 Implementation limitations and conclusion

All three mandatory components demonstrate the principal mechanisms expected in this educational lab: probabilistic PoW mining and cumulative-work fork choice; Casper FFG-style checkpoint finality, slashing and inactivity recovery; and eventual network convergence after partition healing.

The optional experiments further demonstrate quantitative double-spend risk, reusable mining-worker performance and minority selfish-mining incentives. E1 and E3 reproduce their target analytical behaviors closely. E2 demonstrates worker reuse and one near-linear scaling run, while revealing variability in the timing distribution and a standalone implementation rather than integration into the original miner. All bonus scores remain subject to the course's all-or-nothing criteria.

Reference: S. Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System (2008), Section 11; I. Eyal and E. G. Sirer, Majority Is Not Enough: Bitcoin Mining Is Vulnerable (2014).

Submission repository: https://github.com/korneliasara/blockchain-lab03-consensus-simulation